

DANMARKS TEKNISKE UNIVERSITET



(02417) Time Series Analysis

ASSIGNMENT 1

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1 Plot data

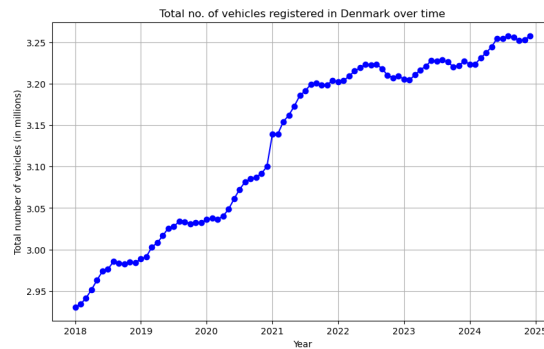


Figure 1: Plot of the total no. of vehicles registered in Denmark from 2018 to 2025

As shown in Figure 1, the total number of registered vehicles in Denmark follows a clear upward trend. From 2018 to 2020, the increase is almost steady, after which a high rise occurs beginning with the first Danish COVID-19 lockdown in March 2020. This high rise continues until late 2022, when COVID-19 restrictions were largely lifted and the disease became managed similarly to seasonal influenza. Although fluctuations are observed thereafter, the overall trend remains positive although slower than earlier.

I don't know if it is obvious that COVID-19 is the reason for these changes in the sudden increase.
In general maybe be vary of concluding on behaviour and just describe what you see.

2 Linear trend model

The matrix form of the model for the first 3 time points in the dataset, $t=1,2,3$ is given by:

$$\mathbf{y} = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix} \quad (1)$$

Here y is an output vector, X the design matrix and θ the parameter vector. For the given dataset, the numerical values of y and X are:

$$\mathbf{y} = \begin{bmatrix} 2.930 \\ 2.934 \\ 2.941 \end{bmatrix} \quad (2)$$

$$\mathbf{X} = \begin{bmatrix} 1 & 2018.000 \\ 1 & 2018.083 \\ 1 & 2018.167 \end{bmatrix}$$

This implies that in January 2018, the total number of vehicles was 2.930 million. In February the number increased to 2.934 million, and in March 2018 it further increased to 2.941 million

3 OLS - global linear trend model

3.1 OLS estimates

The model is given by:

$$Y_t = \theta_1 + \theta_2 x_t + \epsilon_t \quad (3)$$

The parameters were estimated using Ordinary Least Squares (OLS) and are given by

$$\hat{\theta} = (X^\top X)^{-1} X^\top y \quad (4)$$

This is the closed solution for linear regression. We choose the parameters $\hat{\theta}$ that minimize the sum of squared residuals:

$$\sum_{t=1}^n (Y_t - \theta_1 - \theta_2 x_t)^2 \quad (5)$$

The estimates were obtained in R by the function `lm()` which computes the OLS solution.

3.2 Estimation plot

I think in general *or for the last 2 assignments).
They don't care on how you obtain the values, as in what type of code/
language and methods

The results of 3.1 were: $\hat{\theta}$ estimates:

$$\hat{\theta}_1 = -1.104e+02$$

$$\hat{\theta}_2 = 5.614e-02$$

With the standard errors:

$$\hat{\sigma}_{\hat{\theta}_1} = 3.594e+00$$

$$\hat{\sigma}_{\hat{\theta}_2} = 1.778e-03$$

This shows that the model can be explained by:

$$\hat{Y}_t = -110.4 + 0.05614x_t \quad (6)$$

This means that the total number of vehicles increases by $0.05614 \pm 1.96 \cdot 0.001778$ million in a 95% confidence interval, corresponds to roughly to $56,614 \pm 3,490$ vehicles each year in a 95% confidence interval

As in 1 the total number of registered vehicles in Denmark is plotted in the time period 2018 to 2024, the estimated mean is plotted as the red line and represents Y_t :

$$Y_t = \theta_1 + \theta_2 x_t \quad (7)$$

see Figure 2 for representation.

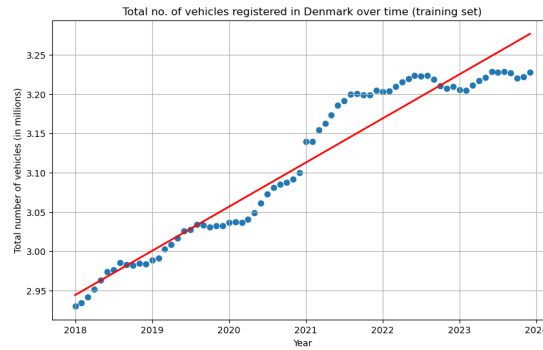


Figure 2: Plot of the total no. of vehicles registered in Denmark from 2018 to 2024, the estimated mean is plotted as the red line

3.3 Forecast

A forecast of the following 12 months (2024-Jan to 2024-Dec.) was obtained by using the fitted linear trend model estimated on the training data.

The forecasts and corresponding 95% prediction intervals were computed in R using the function `predict(..., interval = "prediction")`. This function takes the model found in 3.1 and 3.2 see Formula 7, to compute new observations in the test set, and computes predicted values for each future time point. In the function the using interval = "prediction" the function constructs prediction intervals of the form

$$\hat{Y}_t \pm 1.96 \cdot \hat{\sigma}_{\text{pred}} \quad (8)$$

Where $\hat{\sigma}_{\text{pred}}$ accounts for both uncertainty in the estimated regression parameters and the variance of the future random error term. The prediction interval reflects the range on which a future observation is expected to lie with 95% probability. The results are presented in Table 1.

Month	y_{pred}	Lower 95%	Upper 95%
January 2024	3.281154	3.227579	3.334728
February 2024	3.285832	3.232198	3.339467
March 2024	3.290511	3.236815	3.344208
April 2024	3.295190	3.241430	3.348950
May 2024	3.299869	3.246044	3.353693
June 2024	3.304547	3.250656	3.358439
July 2024	3.309226	3.255267	3.363185
August 2024	3.313905	3.259876	3.367934
September 2024	3.318583	3.264483	3.372683
October 2024	3.323262	3.269090	3.377435
November 2024	3.327941	3.273694	3.382188
December 2024	3.332620	3.278297	3.386942

This is not quite right. You should have used equation (3.61) as it also account for the uncertainty to increase as the prediction gets further and further away from the training set

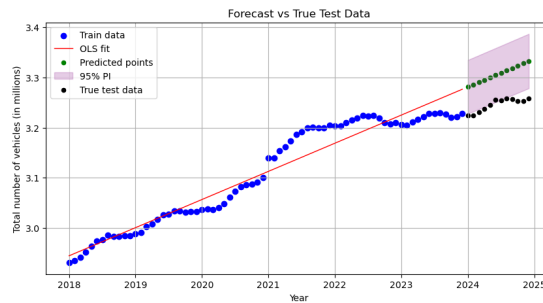
Table 1: Predicted values and 95% confidence intervals for vehicle registrations in Denmark in 2024.

3.4 Forecast plot

The forecast and the corresponding 95% prediction intervals are illustrated in Figure 3

Since the model is a global linear trend model, the forecast increase linearly over time. The monthly increase will approximately correspond to the annual slope divided by 12, reflecting the constant growth implied by the model.

However, comparing the forecasts with the observed test data a clear seasonal pattern can be observed. Similar seasonal patterns are also present in the training data. This shows higher levels during the summer months and an additional increase toward the end of the year. These recurring patterns are not captured by the simple linear trend model.



In this case you cannot notice the difference but suppose the training set ends after 2019, then applying eq (3.61) and not your definition would result in a prediction interval that increases as the test set comes closer and closer to 2025

Figure 3: Plot of the total no. of vehicles registered in Denmark from 2018 to end of 2024. The estimated mean is plotted as the red line. From January 2024 to December 2024 is the estimated forecast represented by the green point and the purple area is 95% prediction intervals. The black points are the true data points.

3.5 Forecast quality

The global linear trend model captures the overall upward trend in the total number of vehicles. The positive slope reflects a steady long-term increase over time as seen in Figure 3.

However the seasonal patterns are not taken into consideration. The presence of systematic seasonal patterns indicates that the residuals are not purely random, which are violating the i.i.d assumption

Furthermore, the rises in the total number of vehicles during COVID-19, explain in Chapter 1, suggests that structural changes are not adequately captured by a simple linear trend model.

Overall, the model provides a reasonable description of the long-term trend. It is however too restrictive to fully describe the underlying dynamics of the time series.



3.6 Residual analysis

Figure 4 shows the residuals plotted against the fitted values. If we have a correctly specified linear trend model, the residuals should be randomly scattered around 0 (the red line) and without systematic pattern.

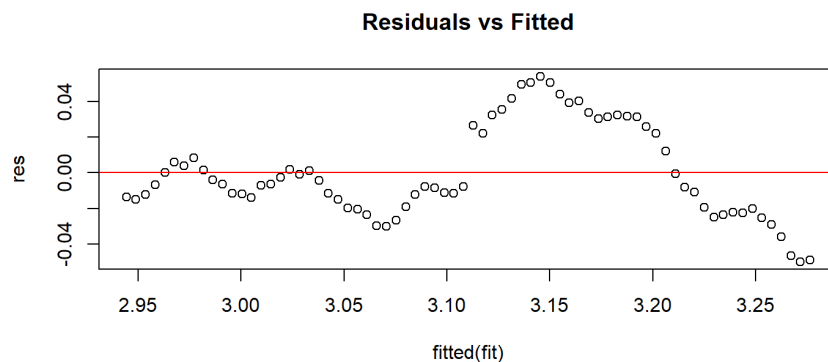


Figure 4: Residuals plotted against the fitted values for the global linear trend model. Each point represents $(\hat{Y}_t, Y_t - \hat{Y}_t)$. The red line indicates zero residual

Figure 4 however clearly reveals a structured pattern, seen by the wave-like pattern. This suggests the presence of the seasonal effect, as mentioned earlier. The model is therefore misspecified and does not fully describe the underlying structure of the data, as it is violating the i.i.d assumption. More appropriate model would likely include seasonal components.

Yes, or other more adaptive features i.e. caring more about recent data rather than the old.

4 WLS - local linear trend model

4.1 Variance-covariance matrix

I think you are mixing a bit what you are comparing in the WLS and OLS. In the OLS you show the variance matrix for the parameters.
In the WLS you show how the weights have an influence on the how much it should consider the past.
Clearly 2 different things and not something to compare.

$$\Sigma_{\text{WLS}} = \begin{bmatrix} 1773.298 & 0 & 0 \\ 0 & 1595.969 & 0 \\ 0 & 0 & 1436.372 \end{bmatrix} \quad (9)$$

$$\Sigma_{\text{OLS}} = \begin{bmatrix} 12.91 & -6.39 \times 10^{-3} \\ -6.39 \times 10^{-3} & 3.16 \times 10^{-6} \end{bmatrix} \quad (10)$$

As seen in Equation 9, we observe a 3×3 submatrix of the full 72×72 variance–covariance matrix used in the WLS estimation. The matrix is diagonal, meaning all off-diagonal elements (covariances) are zero. This indicates that the observations are assumed to be uncorrelated. The diagonal elements (variances) decreases over time, indicating that earlier observations higher uncertainty while more recent observations are assigned smaller variance. This aligns with the WLS estimation putting greater weight on more recent observations.

In comparison, Equation 10 shows the variance–covariance matrix of the OLS parameter estimates (intercept and slope). The variance of the intercept is relatively large (12.91), whereas the variance of the slope is very small (3.16×10^{-6}). This implies that the intercept estimate are considerably more uncertain than that of the slope estimate. As a result of the small variance of the slope, the covariance are also quite small at almost zero.

4.2 λ -weights

Figure 5 show the λ -weights over time. Here, it can be seen that the last time point (2024) is also the one with the highest weight. The weights increases over time as the WLS puts more weight on newer observations and less weight on older observations. This also results in the total sum of the weights being 9.994. An OLS model would in comparison have a total sum of 72 as it treats all observations with equal importance.

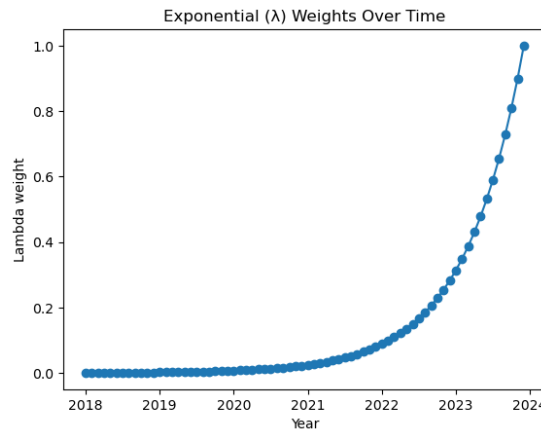


Figure 5: Weights for the training data

4.3 Forecast

The WLS model estimates the values of $\hat{\theta}$ to be

$$\begin{aligned}\hat{\theta}_1 &= -52.4 \\ \hat{\theta}_2 &= 0.02753\end{aligned}$$

With the standard errors being:

$$\begin{aligned}\hat{\sigma}_{\hat{\theta}_1} &= 5.13 \\ \hat{\sigma}_{\hat{\theta}_2} &= 0.00254\end{aligned}$$

This shows that the model can be explained by:

$$\hat{Y}_t = -52.48 + 0.02753x_t \quad (11)$$

This can also be observed in Figure 6, which shows the WLS estimated model together with the 95% prediction interval (PI). The plot shows that the PI captures nearly all future test observations for the period 2024–2025. The predicted points are also quite close to the true test data but as our model isn't linear, it can't capture the small fluctuations in the true data set. Comparing the OLS and WLS predictions, the WLS model appears preferable. The WLS prediction interval contains the future test data within the 95% confidence bounds, whereas the OLS predictions and their corresponding prediction intervals (see Figure 3) fail to capture the true test observations. This suggests that the WLS model provides a better fit for the data and holds better predictive abilities for the dataset.

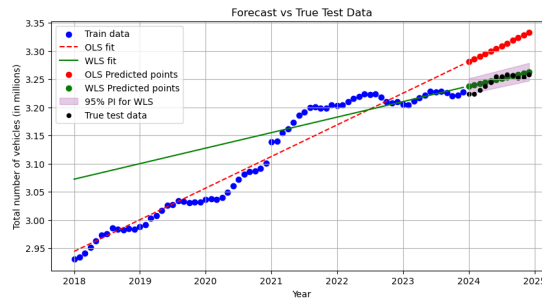


Figure 6: Forecast using the WLS model

It looks like you create the t-dist over the entire set and not over the memory as we have to do when doing WLS prediction, as we don't use information past our memory in the predictions. I am referring to eq. (3.92) and from week 3 slides on WLS and prediction

5 Recursive estimation and optimization of λ

5.1

The first 2 calculated iterations, until the value of R_2 can be found in Figure 7

$$K_t = \frac{R_{t-1} x_t}{\lambda + x_t^T R_{t-1} x_t} \quad \hat{\theta}_t = \hat{\theta}_{t-1} + K_t (y_t - x_t^T \hat{\theta}_{t-1})$$

$$R_t = \frac{1}{\lambda} (R_{t-1} - K_t x_t^T R_{t-1})$$

$y_1 = 2.930483, y_2 = 2.934044, x_1 = [2018], x_2 = [2018, 493]$
 $\hat{\theta}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, R_0 = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}, \lambda = 0.9$

$t=1:$
 $K_1 = \frac{R_0 x_1}{\lambda + x_1^T R_0 x_1} = \frac{\begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} \begin{bmatrix} 2018 \\ 0 \end{bmatrix}}{0.9 + \begin{bmatrix} 2018 & 0 \end{bmatrix} \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} \begin{bmatrix} 2018 \\ 0 \end{bmatrix}} = \frac{\begin{bmatrix} 201.8 \\ 0 \end{bmatrix}}{407233.4} = \begin{bmatrix} 4.955 \cdot 10^{-7} \\ 0 \end{bmatrix}$
 $\hat{\theta}_1 = \hat{\theta}_0 + K_1 (y_1 - x_1^T \hat{\theta}_0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 4.955 \cdot 10^{-7} \\ 0 \end{bmatrix} \cdot (2.930483 - [2018] \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix}) = \begin{bmatrix} 7.19 \cdot 10^{-7} \\ 0.001453 \end{bmatrix}$
 $R_1 = \frac{1}{0.9} \cdot \left(\begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} - \begin{bmatrix} 4.955 \cdot 10^{-7} \\ 0 \end{bmatrix} \begin{bmatrix} 2018 & 0 \end{bmatrix} \right) = \begin{bmatrix} 0.1111 & -5.51 \cdot 10^{-5} \\ -5.51 \cdot 10^{-5} & 2.72 \cdot 10^{-8} \end{bmatrix}$

$t=2:$
 Prediction error: $\hat{y}_2 = x_2^T \hat{\theta}_1 = 1.719 \cdot 10^{-7} + 2018.083 \cdot 0.001453 \approx 2.9326$
 $e_2 = y_2 - \hat{y}_2 = 0.001453$
 $K_2 = \frac{R_1 x_2}{\lambda + x_2^T R_1 x_2} = \frac{\begin{bmatrix} 0.1111 & -5.51 \cdot 10^{-5} \\ -5.51 \cdot 10^{-5} & 2.72 \cdot 10^{-8} \end{bmatrix} \begin{bmatrix} 2018 \\ 493 \end{bmatrix}}{0.9 + \begin{bmatrix} 2018 & 493 \end{bmatrix} \begin{bmatrix} 0.1111 & -5.51 \cdot 10^{-5} \\ -5.51 \cdot 10^{-5} & 2.72 \cdot 10^{-8} \end{bmatrix} \begin{bmatrix} 2018 \\ 493 \end{bmatrix}} = \begin{bmatrix} 0.001233 \\ -6.77 \cdot 10^{-5} \end{bmatrix}$
 $\hat{\theta}_2 = \hat{\theta}_1 + K_2 e_2 = \begin{bmatrix} 7.19 \cdot 10^{-7} \\ 0.001453 \end{bmatrix} + \begin{bmatrix} 0.001233 \\ -6.77 \cdot 10^{-5} \end{bmatrix} \cdot 0.001453 = \begin{bmatrix} 0.002468 \\ 0.001453 \end{bmatrix}$
 $R_2 = \frac{1}{\lambda} (R_1 - K_2 x_2^T R_1) = \begin{bmatrix} 0.0952 & -4.74 \cdot 10^{-5} \\ -4.74 \cdot 10^{-5} & 2.64 \cdot 10^{-8} \end{bmatrix}$

Results:
 $\hat{\theta}_1 = \begin{bmatrix} 7.19 \cdot 10^{-7} \\ 0.001453 \end{bmatrix}, R_1 = \begin{bmatrix} 0.1111 & -5.51 \cdot 10^{-5} \\ -5.51 \cdot 10^{-5} & 2.72 \cdot 10^{-8} \end{bmatrix}$
 $\hat{\theta}_2 = \begin{bmatrix} 0.002468 \\ 0.001453 \end{bmatrix}, R_2 = \begin{bmatrix} 0.0952 & -4.74 \cdot 10^{-5} \\ -4.74 \cdot 10^{-5} & 2.64 \cdot 10^{-8} \end{bmatrix}$

Figure 7: The update equations of R_t and $\hat{\theta}_t$ written on paper

5.2

The updated values for θ_2 and θ_3 are as follows

$$\theta_2 = \begin{bmatrix} -83.30 \\ 0.04 \end{bmatrix}$$

$$\theta_3 = \begin{bmatrix} -129.52 \\ 0.07 \end{bmatrix}$$

We don't get the exact same values in 5.1, but I can't if you or we have done a mistake. But what does NOT make sense is that your θ_2 by hand does not return the same values as the one you have done on the computer. They should have been equal.

5.3

The two estimates can be seen below, where it can be seen that they are close to each other, but they can get even closer by doing some small modifications to our initial guess and certainty matrix.

Method	Intercept ($\hat{\beta}_0$)	Slope ($\hat{\beta}_1$)
OLS (all data)	-97.281935	0.049673
RLS ($t = N, \lambda = 1$)	-96.136460	0.049106
Difference	1.15×10^0	5.67×10^{-4}

Table 2: Comparison of OLS and RLS Estimates

To minimize the difference between RLS and OLS, we set R_{init} to a near-zero value, in our case, $1e^{-9}$. This represents a state of maximum uncertainty, ensuring that the initial guess $\theta_{init} = [0, 0]$ does not bias the final results. As a result, the model 'snaps' to the data-driven OLS estimate almost instantly as the first observations arrive

Table 3: RLS Estimates and Differences

Category	Intercept	Slope
OLS (all data)	-97.281935	0.049673
RLS Estimates ($t = N, \lambda = 1$)	-97.280776	0.049672
Difference	1.16×10^{-3}	5.73×10^{-7}

5.4

Below there are two plots, the plots on Figure 8 we can see the change of the slope and intercept as we increase the time steps. It is clear to see that the 0.99 lambda value makes the model less susceptible to changes over time compared to the 0.7 lambda values which fluctuates a lot.



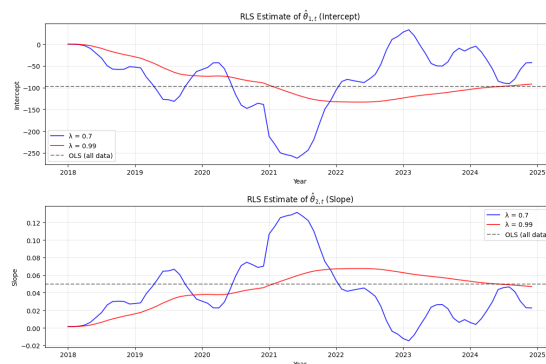


Figure 8: Overview of the intercept and slope of the RLS estimates and how they change over time with different values of λ

5.5

Looking the residuals on Figure 9, we can see that both forecasts have a tendency to underestimate the values at the beginning while increasingly overestimating at the end. The $\lambda = 0.7$ has generally a smaller band because the forgetting factor allows it to more easily react to sudden drops because of it of having less weight on earlier observations. Unlike $\lambda = 0.99$ where we can see that the model is still heavily biased by early observations and takes longer to react.

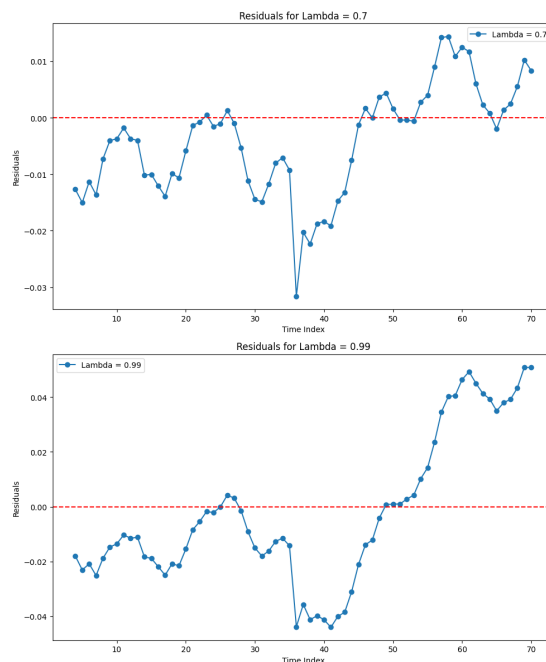


Figure 9: Overview of the residuals of the RLS estimates and how they change over time with different values of λ

5.6

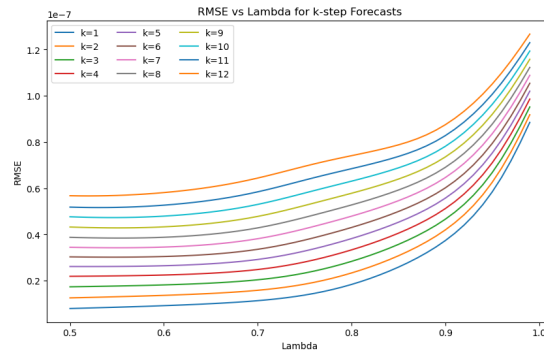


Figure 10: Plot of RMSE given different horizons and λ -values

On Figure 10 we can see the RMSE plotted for λ values from 0.5 to 0.99 for k -step horizons from 1 to 12. The results show a consistent pattern across all horizons. RMSE is lowest for $k = 1$ and increases as k grows, with $k = 12$ exhibiting the highest RMSE. This is expected, since larger values of k correspond to longer forecast horizons, making prediction more difficult and therefore increasing forecast error. For all horizons, the RMSE increases as λ increases, and the growth appears increasingly exponential for larger values of λ .

To choose an optimal value of λ , we would select the value that minimises the average RMSE across all horizons. This ensures that the forgetting factor balances performance over both short- and long-term forecasts. This should result in a simple and robust forgetting factor. An alternative approach would be to let λ depend on the forecast horizon, using different forgetting factors for different values of k . This could potentially improve performance by faster adapting to short horizons and being more stable for long horizons. However, we would still choose a single λ as it avoids additional complexity.

It depends, maybe you would want 2 models.
1 that predict 1 year ahead and the other 1 month ahead.
Then you can just have 2 models for the same data .

5.7

For Figure 11 we have chosen the value of λ that minimises the average RMSE, which was 0.52. The plot shows that the RLS predictions are very close to the WLS predictions, indicating similar forecasting performance. The 95% prediction interval for RLS covers all of the true test observations. This suggests that RLS has a reasonable level of uncertainty and that it performs comparably to WLS. In contrast, the OLS predictions deviate substantially from the observed test data, with almost none of the OLS predictions lying within the RLS prediction interval. This shows that RLS (and WLS) are better at forecasting compared to OLS for this dataset.

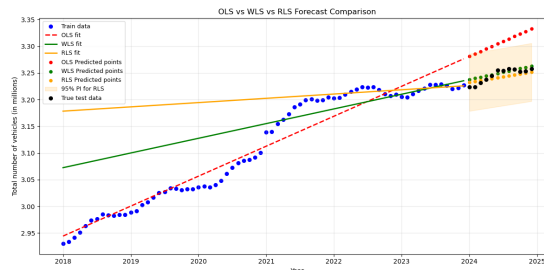


Figure 11: Forecast comparison of OLS, WLS and RLS

5.8 Pitfalls

Challenges that arise with tests set is that they don't consider big changes in the environment that can make a big difference for what the next time step will be. Although big changes can occur, generally recursive estimations do provide good tools to at least make a grounded forecast in data with time dependent steps.